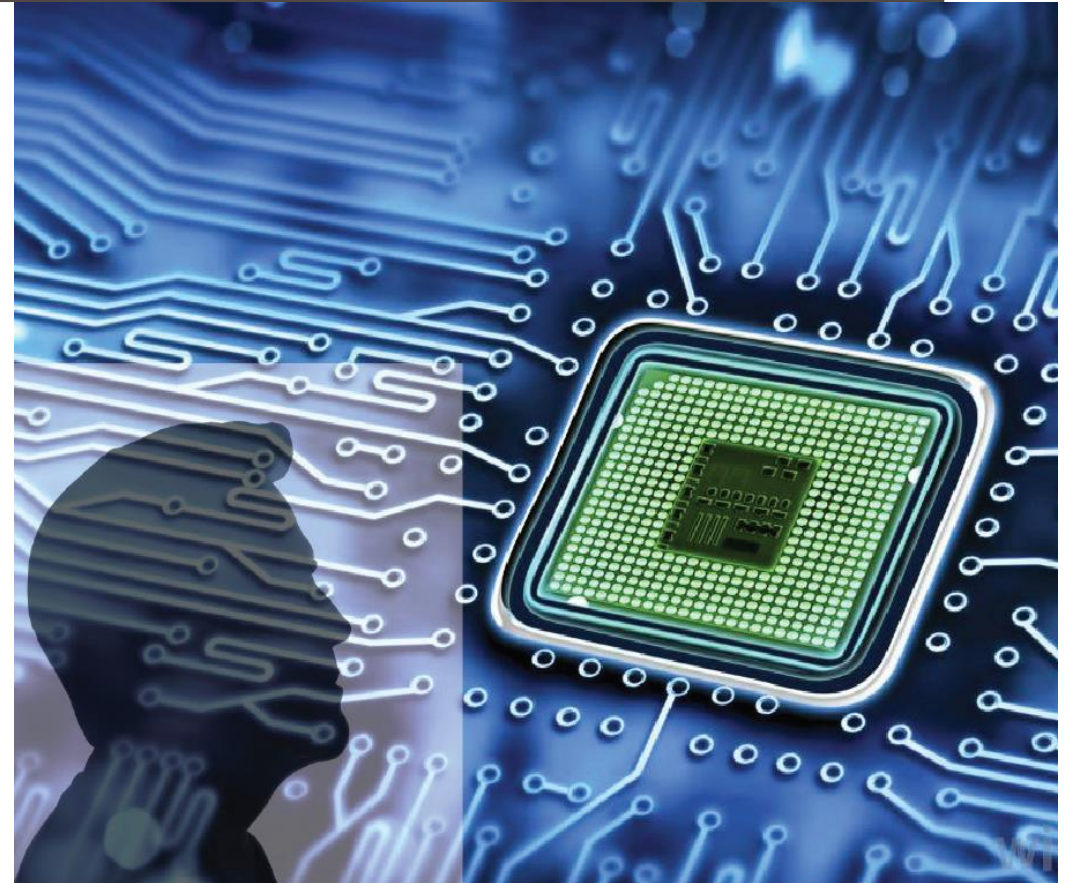


VLSI DESIGN FLOW

Week-8 Lecture-1 STA

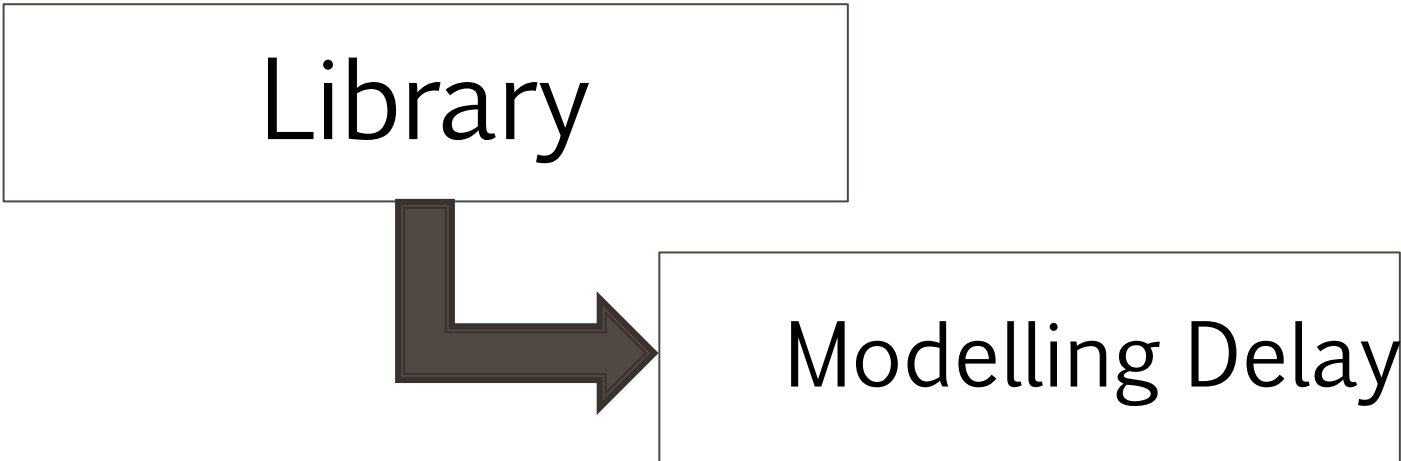
Sneh Saurabh
30th September, 2024



VDF: Revision/Announcements

-



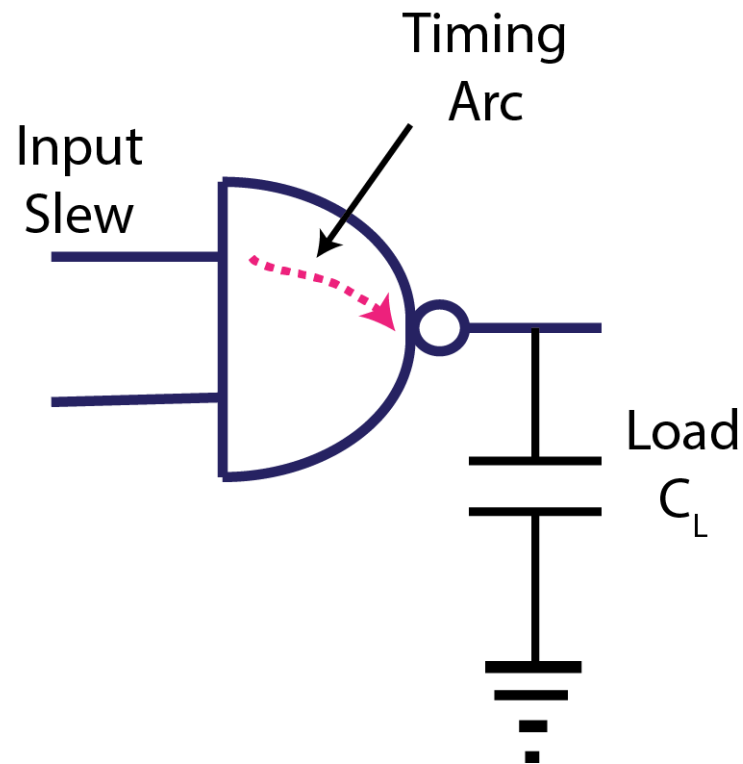


```
graph TD; Library[Library] --> ModellingDelay[Modelling Delay];
```

Library

Modelling Delay

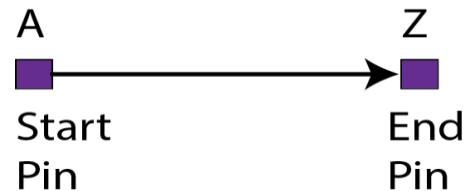
CMOS : Characteristics of Slew and Delay



- In general, the *delay* (D) and *output slew* (S_{OUT}) of a given *timing arc* depend on:
 - Input Slew (S_{IN}): Slew of the signal at the input pin of the given arc
 - Output Load (C_L): Capacitance driven by the output pin of the given arc

- The relationship may be non-linear:
 - $D = f(S_{IN}, C_L)$
 - $S_{OUT} = g(S_{IN}, C_L)$
 - f, g are non-linear function, but are typically monotonically increasing with S_{IN} and C_L

Non-linear Delay Model (NLDM)

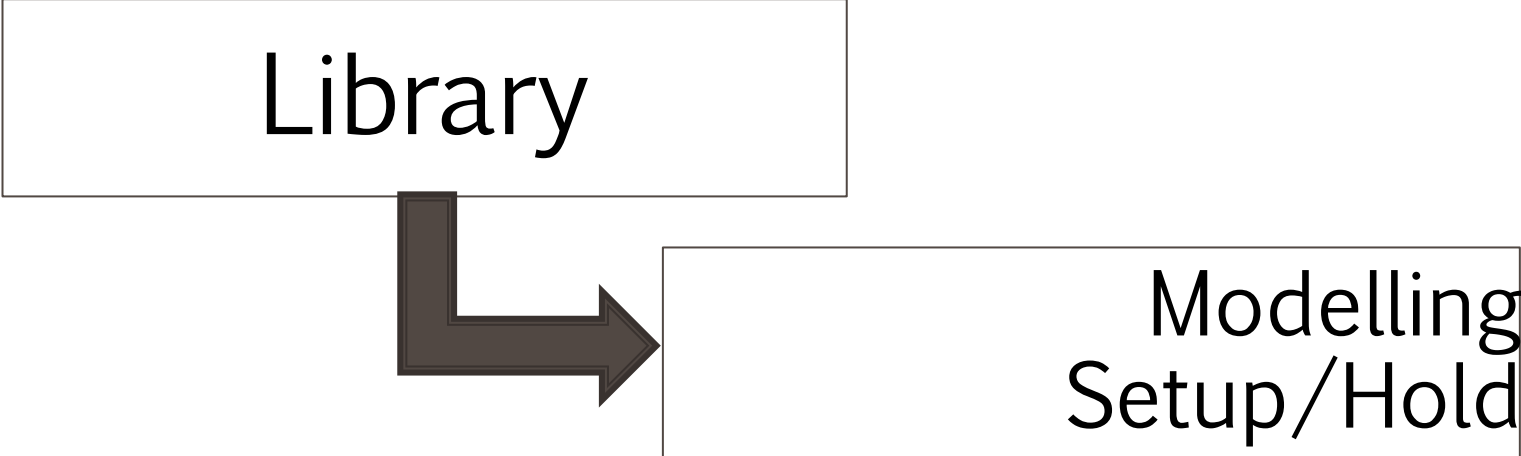


- In library, the *delay* (D) and the *output slew* (S_{OUT}) are modelled as two dimensional discrete point tables
 - There are different tables for delay and output-slew
-
- For each combination of S_{IN} and C_L , the *delay* (D) and the *output slew* (S_{OUT}) are defined in the corresponding tables
 - Intermediate values are typically *linearly interpolated* from closest match

```
u_table_template(index_1) {  
    variable_1 : input_net_transition ;  
    variable_2 : total_output_net_capacitance ;  
    index_1( "10, 20, 30" ) ;  
    index_2( "1.2, 5.0, 15.0, 37.5" ) ;  
}  
...  
pin(Z) {  
    timing() {  
        related_pin      : "A" ;  
        timing_sense     : positive_unate ;  
        cell_rise(index_1) {  
            values( " 4, 5, 7, 12,  
...3x4 table);  
        }  
        ...  
    }  
}
```

Library : Advanced Delay Model

- At advanced process nodes simple NLDM model is not accurate
- Other delay models based on current source model are employed
 - Composite Current Source (CCS)
 - Effective Current Source Model (ECSM)

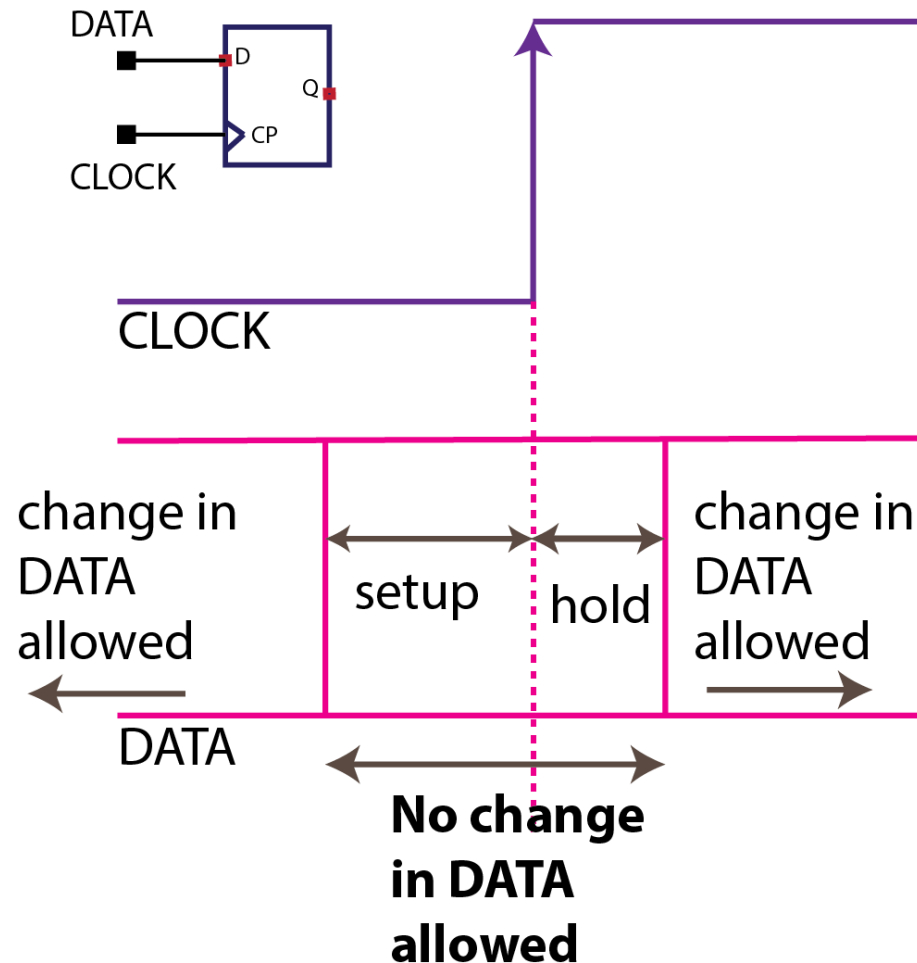


```
graph TD; Library[Library] --> Modelling[Modelling Setup/Hold];
```

Library

Modelling
Setup/Hold

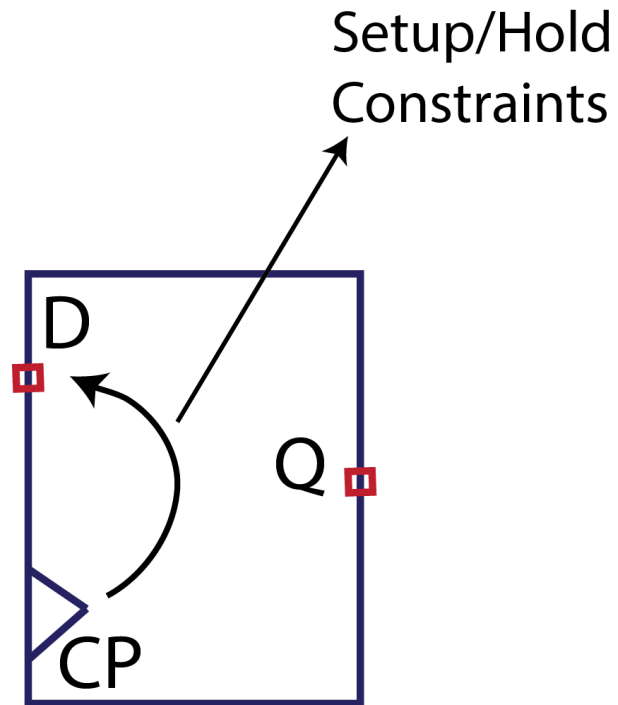
Setup/Hold Time : Definition



- **Setup time:** minimum amount of time the DATA signal should be held steady **before the CLOCK edge** so that the DATA is sampled correctly and deterministically

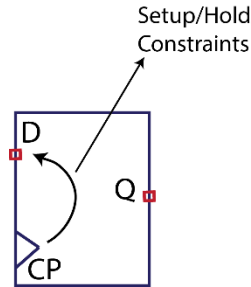
- **Hold time:** the minimum amount of time the DATA signal should be held steady **after the CLOCK edge** so that the DATA is sampled correctly and deterministically

CMOS : Setup/Hold Characteristics



- In general, the *setup* (SU) and *hold* (H) constraints of a given sequential elements depend on:
 - Data Slew (S_D): Slew of the signal at the data pin 'D' pin of the sequential element
 - Clock Slew (S_{CLK}): Slew of the clock signal at the clock pin 'CP' pin of the sequential element
- The relationship may be non-linear:
 - $SU = f(S_D, S_{CLK})$
 - $H = g(S_D, S_{CLK})$
 - f, g are non-linear functions

Library : Modelling Setup/Hold Constraints



- In library, the **setup** (SU) and the **hold** (H) constraints are modelled as two dimensional discrete point tables
- There are different tables for setup and hold
- For each combination of S_D and S_{CLK} , the **setup** (SU) and the **hold** (H) constraints are defined in the corresponding tables
- Intermediate values are typically **linearly interpolated** from closest match

```
u_table_template(index_1) {  
    variable_1 : constrained_pin_transition;  
    variable_2 : related_pin_transition;  
    index_1( "10, 20, 30" );  
    index_2( "10, 20, 30, 40" );  
}  
...  
pin(D) {  
    timing() {  
        related_pin      : "CP" ;  
        timing_type : "setup_rising";  
        rise_constraint(index_1) {  
            values( " 4, 5, 7, 12,  
...3x4 table);  
        }  
        ...  
    }  
}
```

Library : Other information contained in library

- Power models
- Models for crosstalk noise analysis
- Power/Ground Pin Information
- State dependent arcs: *sdf_cond*, *when*
- *Wire Load Models*
- Other attributes that may be vendor specific

Static Timing Analysis



Basics

Static Timing Analysis (STA): What is done?

- Ensures that the circuit is in a valid state at each clock cycle
- Verifies that the design is capable of operating at the *given frequency*
 - Information of *frequency* comes from *constraints*
- Ensures that the design does not have *setup* or *hold* violation at flip-flop

- The analysis is based on *worst case* scenario and takes a pessimistic view wherever possible
 - Verification is done without test vector and simulation (therefore **static**)
 - Ensures that design will not have setup or hold violations for **any test vector**

STA

Simulation

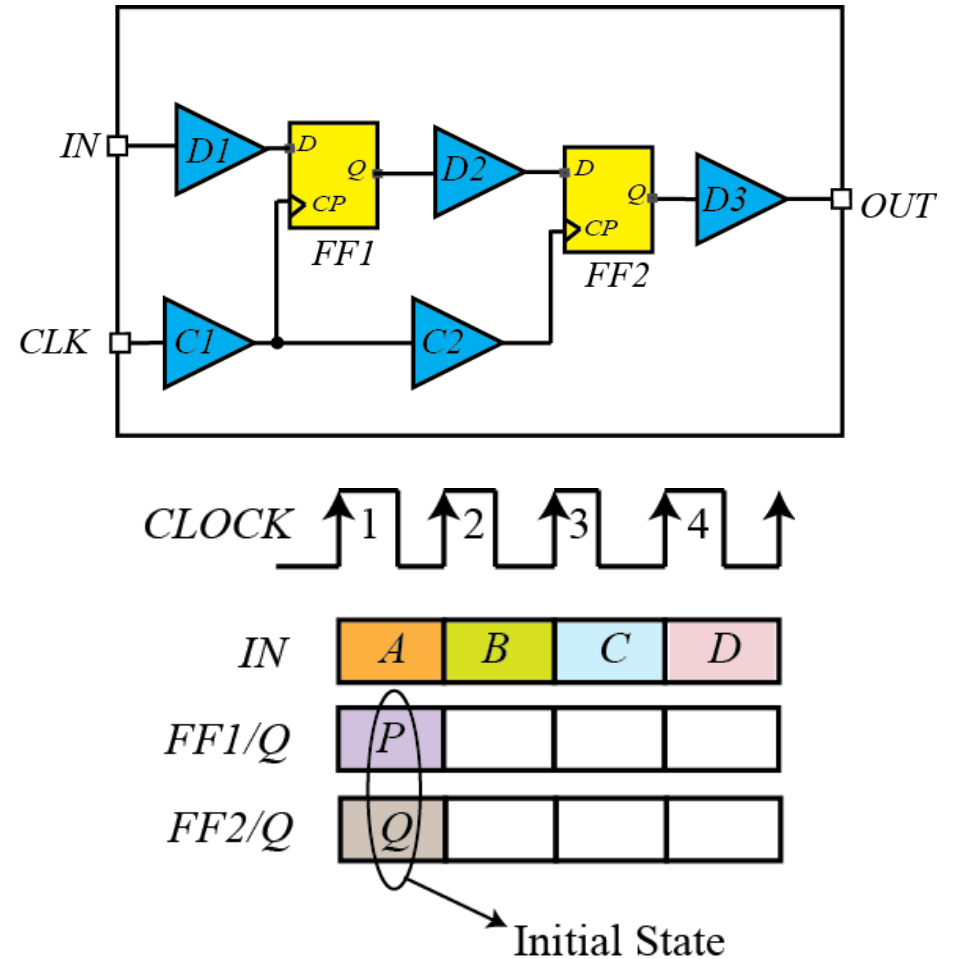
Synchronous Circuit: Data Propagation

Consider a synchronous circuit shown alongside.

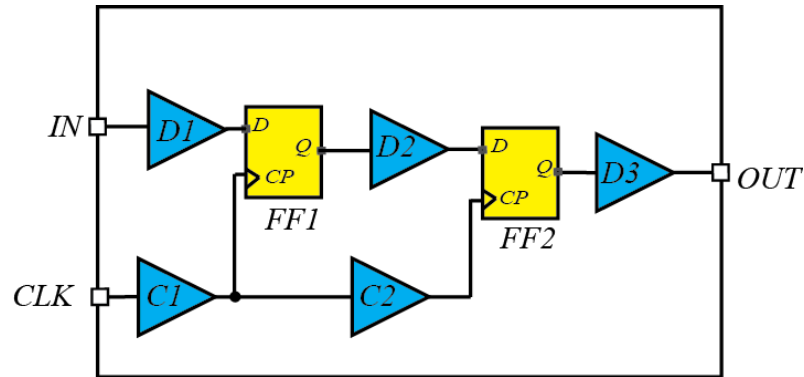
Assume that:

- Flip-flops are ideal
- Buffers have some delay
- Inputs at the port IN are applied as shown [using identifiers for clarity]
- State defined by the combination of values at the Q-pin
- Initial state is {PQ}

Let us understand the behavior of the circuit in different clock cycles for various cases of delay of the buffers.

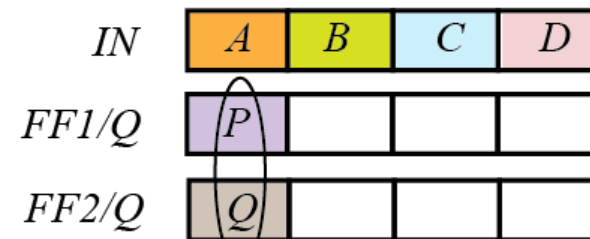
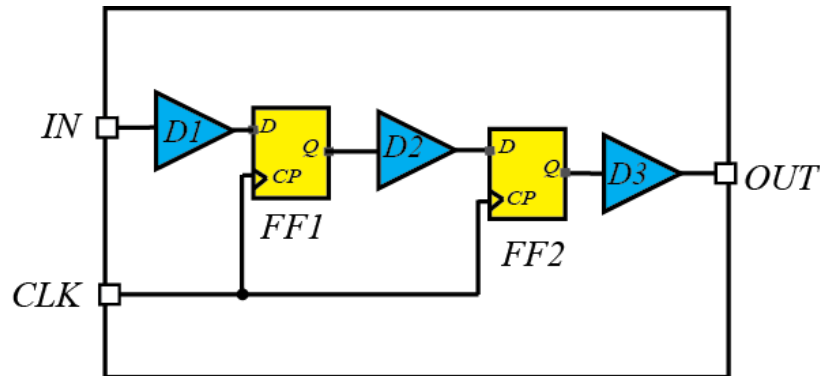


Synchronous Circuit: Synchronous Behavior



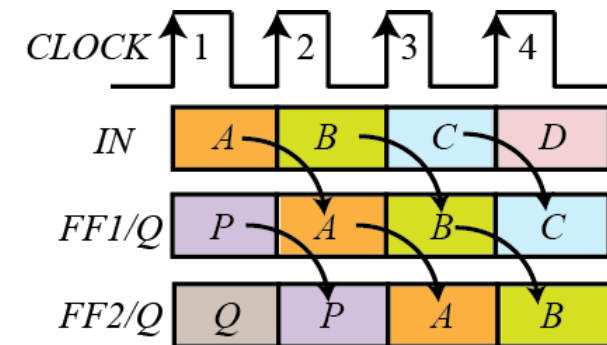
Assume that:

- Delay of $D1$, $D2$, and $D3$ be some finite value less than the clock period, and
- Delays of $C1$ and $C2$ are negligible.



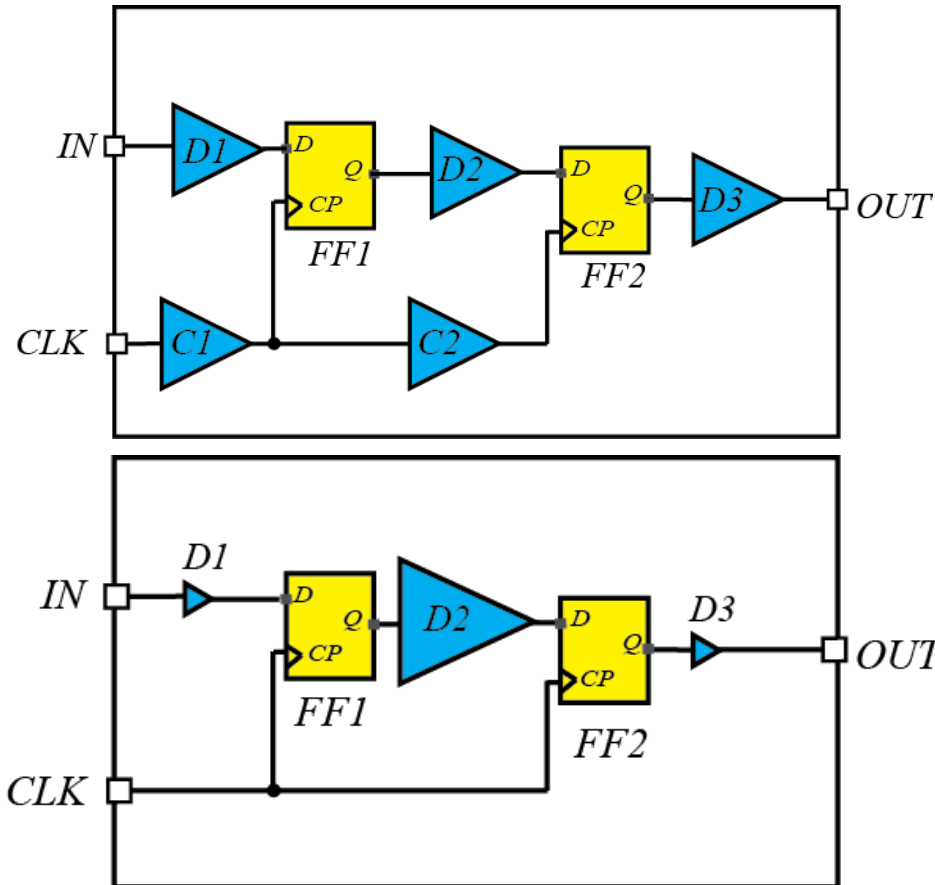
Initial State

Cycle	State
1	{PQ}
2	{AP}
3	{BA}
4	{CB}



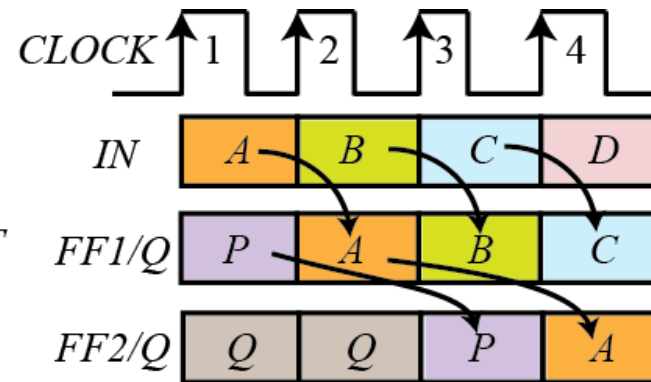
These are valid states of a synchronous circuit.

Synchronous Circuit: Zero Clocking



Assume that:

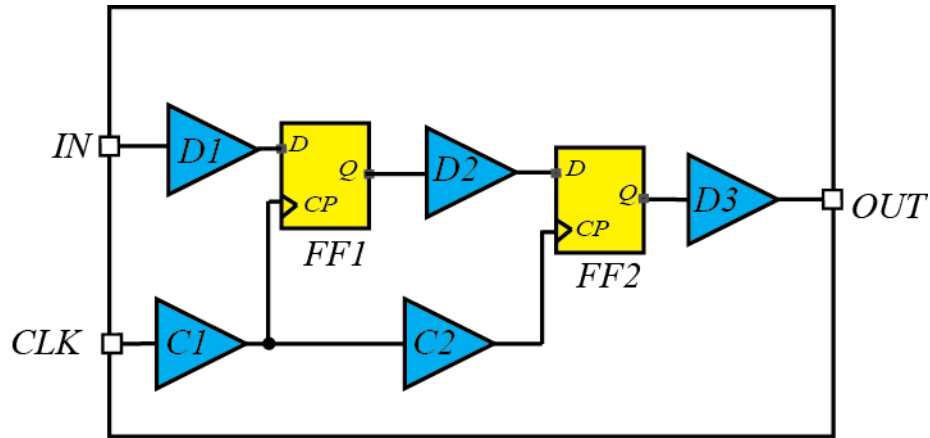
- Delay of $C1$ and $C2$ are negligible.
- Delay of the circuit element $D2$ is more than one clock period, but less than two clock periods.
- Delay of $D1$ and $D3$ are minimal.



Cycle	State	Valid State
1	{PQ}	{PQ}
2	{AQ}	{AP}
3	{BP}	{BA}
4	{CA}	{CB}

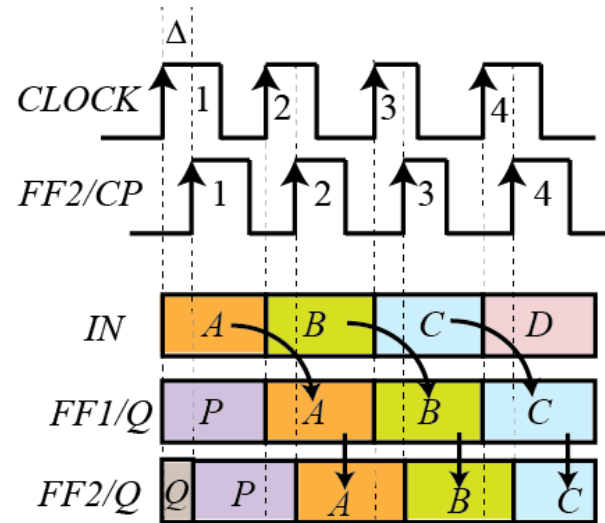
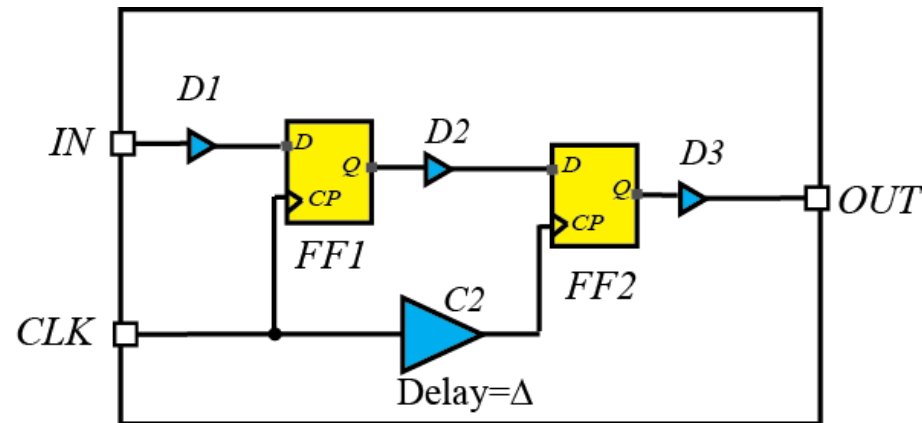
- In effect clock fails to capture the right data (zero-clocking) due to late data arrival
- The circuit goes into invalid states

Synchronous Circuit: Double Clocking



Assume that:

- Delay of circuit elements $D1$, $D2$, $D3$, and $C1$ are insignificant.
- Delay of the circuit element $C2$ is Δ



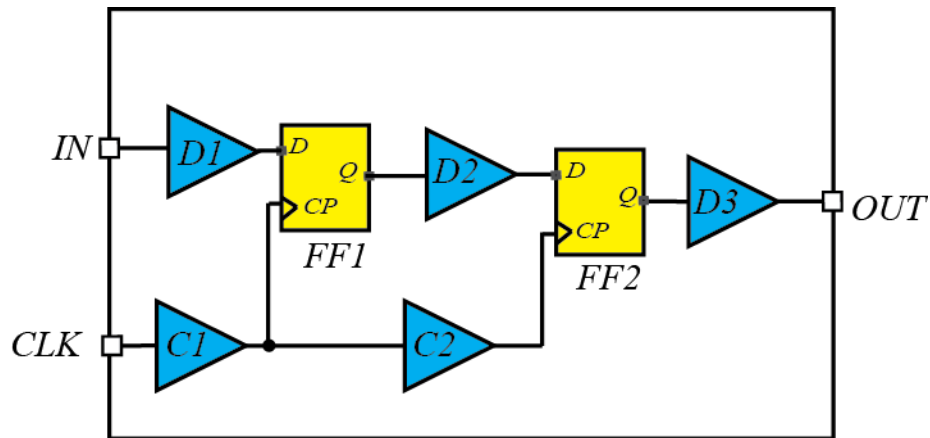
Cycle	State	Valid State
1	{PP}	{PQ}
2	{AA}	{AP}
3	{BB}	{BA}
4	{CC}	{CB}

- In effect, the data gets captured by two flop-flops by the same clock edge (double clocking)
- The circuit goes into invalid states

Synchronous Circuit: Verification

To ensure synchronous behavior:

- **Avoid Zero Clocking:** setup analysis or late analysis
- **Avoid Double Clocking:** hold analysis or early analysis
- A synchronous circuit can contain many flip-flops
 - Data can propagate sequentially through a pipeline before reaching the output
- Examine each pair of launch and capture flip-flops separately

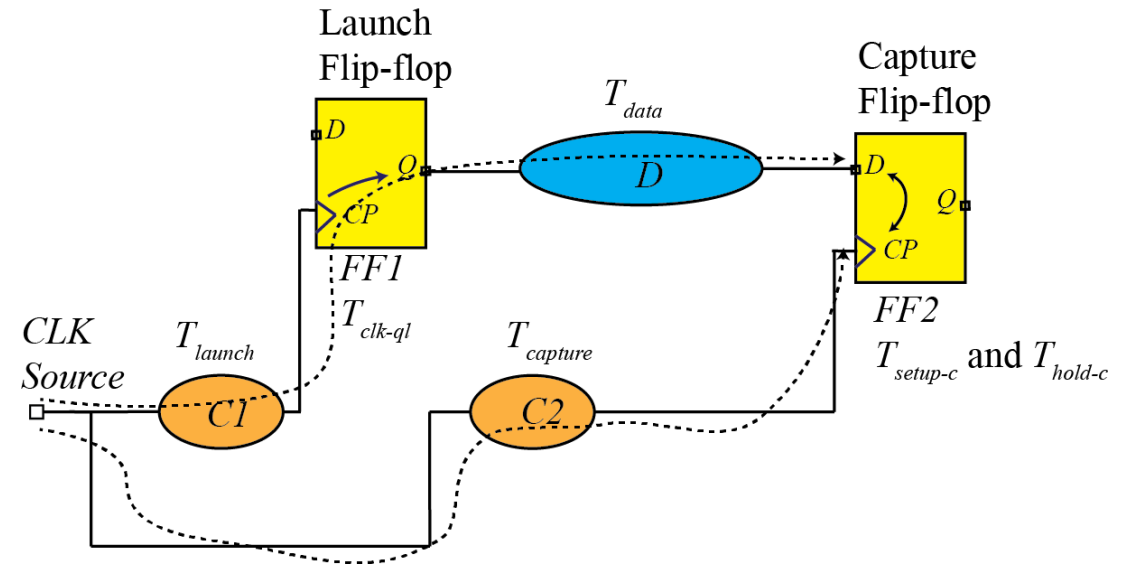


- Various types of combinational gates can be encountered in a path
- Add the delay of all the combinational circuit elements (and also the wire delay) in the path and check for delay requirements

- Real flip-flops have ST, HT, CK-Q delay
- Account for them (make the verification a bit more pessimistic)

Static Timing Analysis (STA) : Setup Requirement (1)

- Ensures that the data sent by launch flip-flop in a given clock cycle is captured reliably by the capture flip-flop in *the next clock cycle*
- Ensures that the setup requirement of the flip-flop is also met



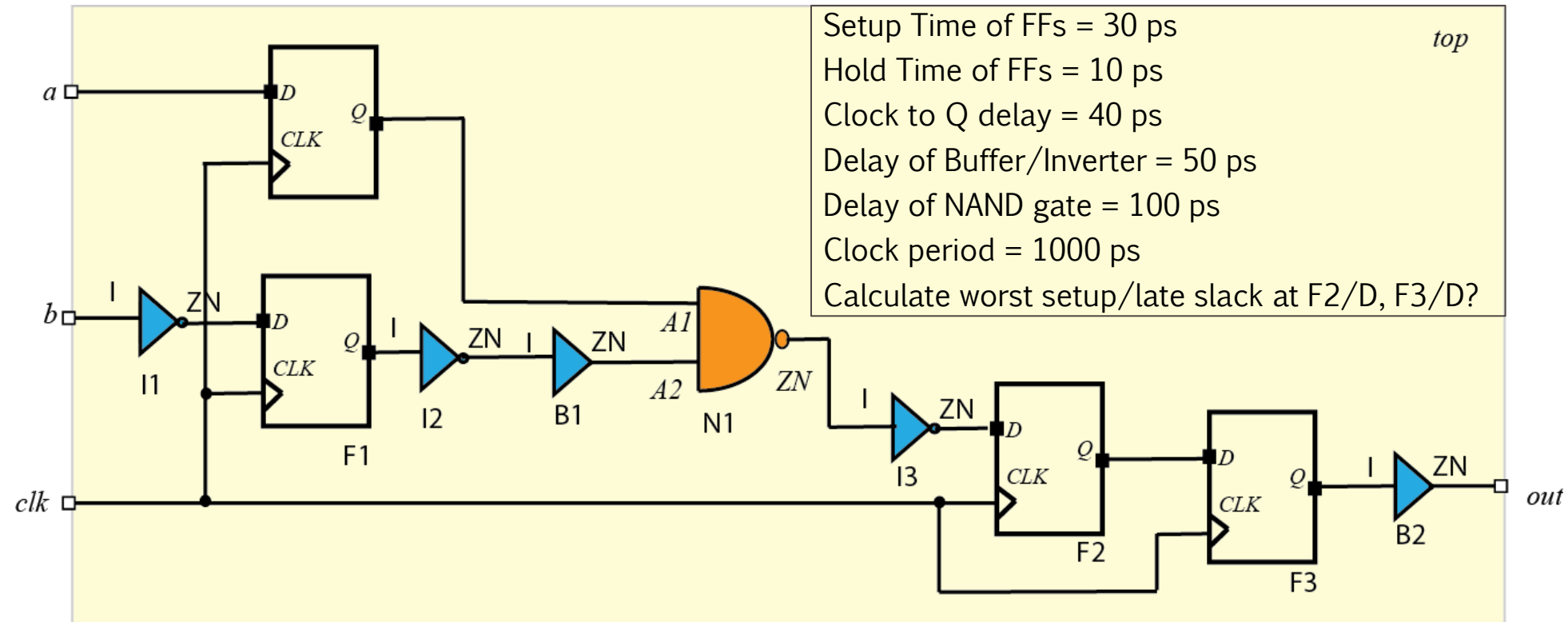
Arrival Time of data at the D-pin: $t_{arrival} = T_{launch} + T_{clk-ql} + T_{data}$

Required time for data to settle at FF2/D: $t_{req,set} = T_{period} + T_{capture} - T_{setup-c}$

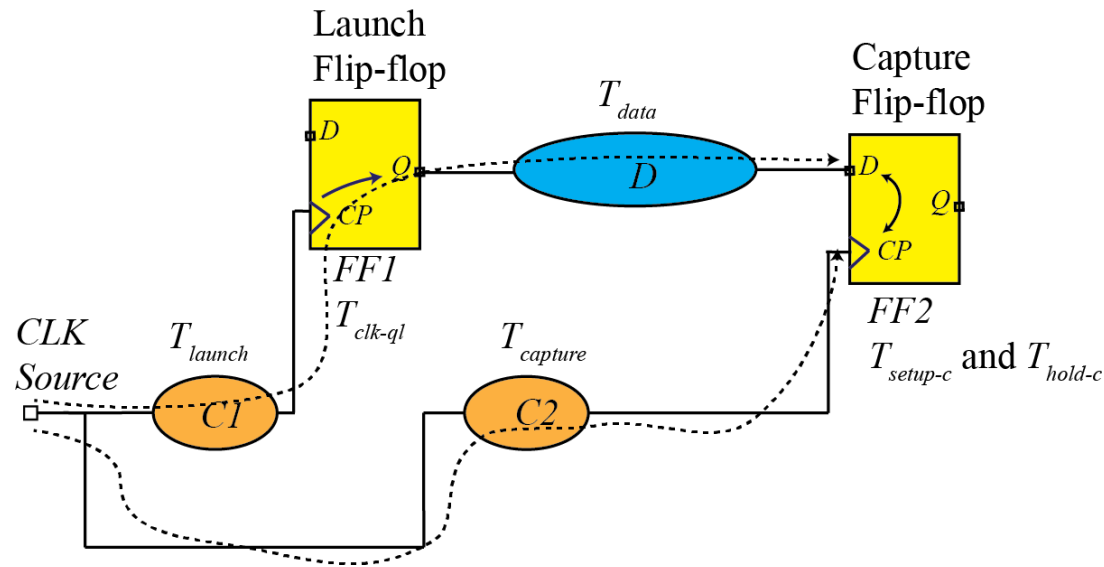
To avoid zero clocking and setup-time constraints of flip-flops: $t_{req,set} > t_{arrival}$

Setup requirement: $T_{period} + T_{capture} - T_{setup-c} > T_{launch} + T_{clk-ql} + T_{data}$

Static Timing Analysis (STA) : Problem



Static Timing Analysis (STA) : Setup Requirement (2)



Setup requirement:

$$T_{period} + T_{capture} - T_{setup-c} > T_{launch} + T_{clk-ql} + T_{data}$$

$$T_{period} > (T_{launch} - T_{capture}) + T_{clk-ql} + T_{data} + T_{setup-c}$$

$$T_{period} > \delta_{lc} + T_{clk-ql} + T_{data} + T_{setup-c} \quad [\delta_{lc} \text{ is the clock skew}]$$

Most Restrictive:

$$T_{period} > \delta_{lc} + T_{clk-ql} + T_{data,max} + T_{setup-c}$$

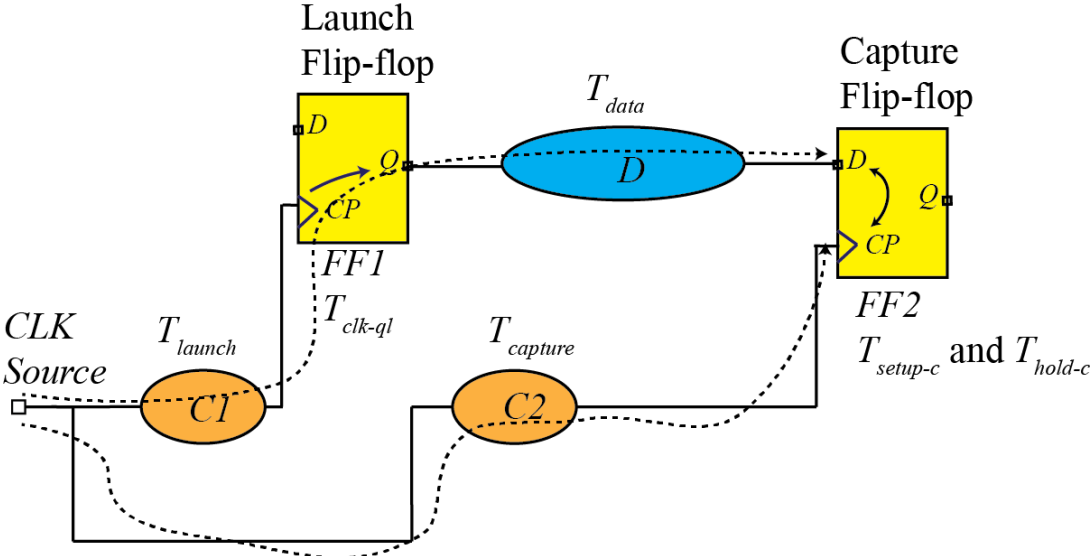
Setup Violations can occur if:

- Clock Period is decreased (clock frequency is increased)
- Delay of capture clock path is decreased
- Delay of data path is increased
- Delay of launch clock path is increased

What happens for an ideal flip-flop?

What happens for an ideal clocking structure?

- Ensures that the hold requirement of the flip-flop is also met



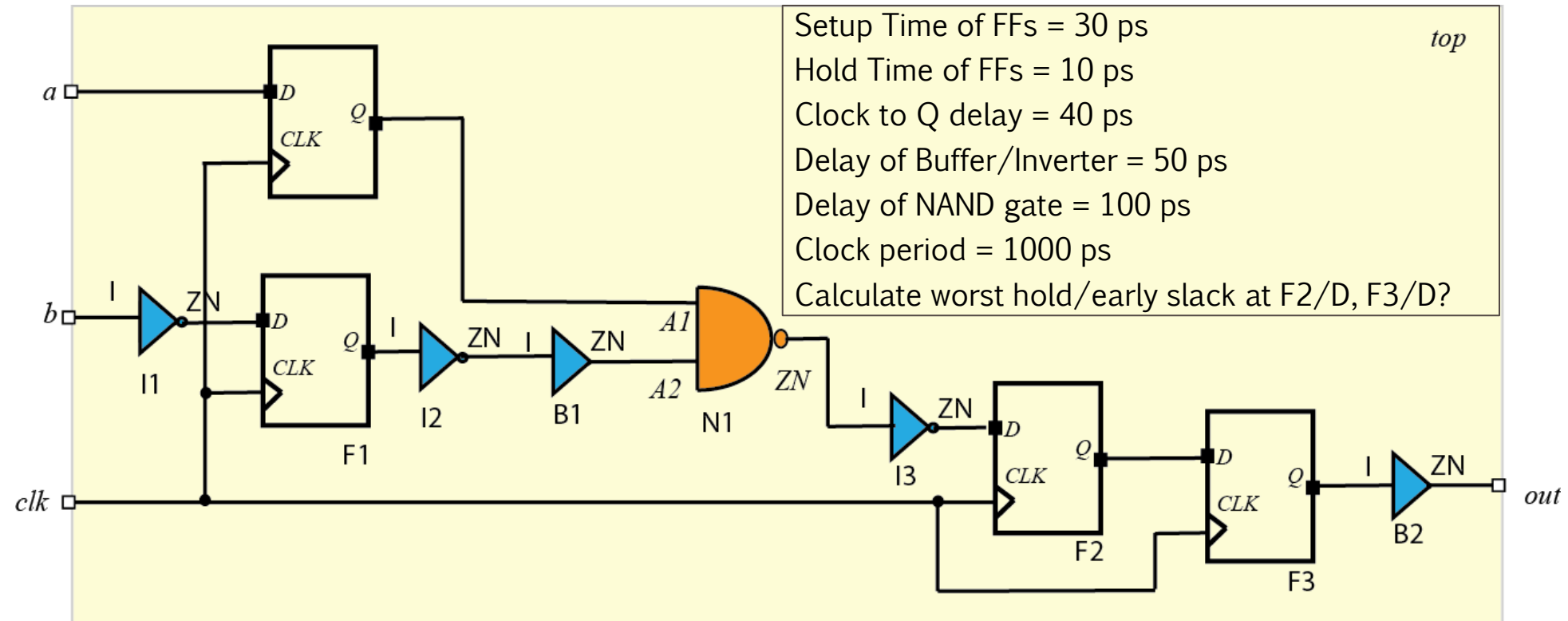
Data Reaches D-pin: $t_{arrival} = T_{launch} + T_{clk-ql} + T_{data}$

Data to arrive at FF2/D after the required time: $t_{req,hold} = T_{capture} + T_{hold-c}$

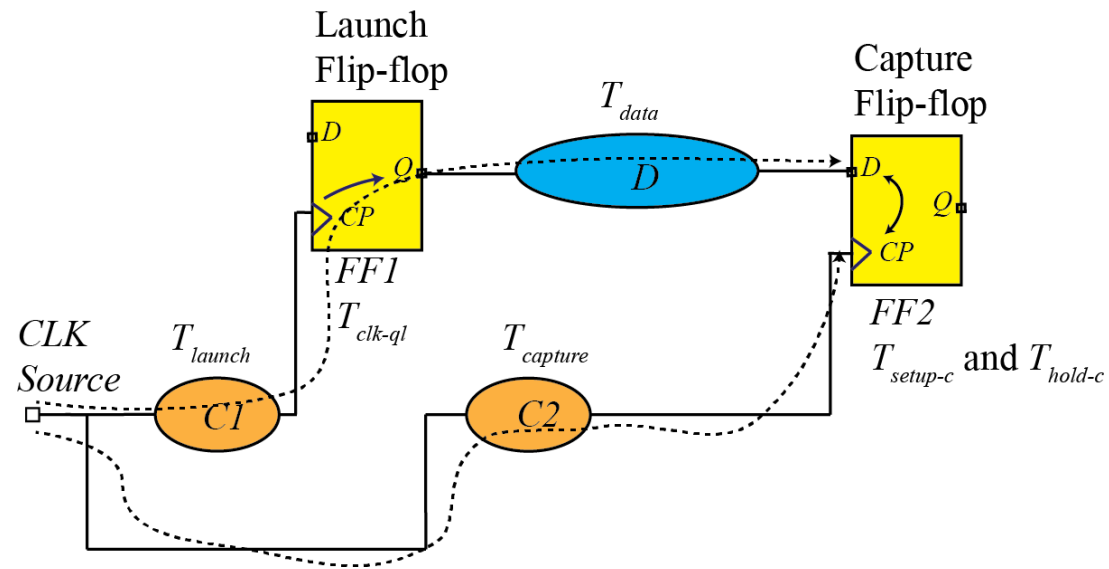
To avoid double clocking and hold-time constraints of flip-flops: $t_{arrival} > t_{req,hold}$

Hold requirement: $T_{launch} + T_{clk-ql} + T_{data} > T_{capture} + T_{hold-c}$

Static Timing Analysis (STA) : Problem



Static Timing Analysis (STA) : Hold Requirement (2)



Hold requirement:

$$T_{launch} + T_{clk-ql} + T_{data} > T_{capture} + T_{hold-c}$$
$$\delta_{lc} + T_{clk-ql} + T_{data} > T_{hold-c}$$

Most Restrictive requirement:

$$\delta_{lc} + T_{clk-ql} + T_{data,min} > T_{hold-c}$$

- Hold Violations can occur if:
 - Delay of data path is decreased
 - Delay of launch clock path is decreased
 - Delay of capture clock path is increased

References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.